

WASP-12b

R-1592

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_191117 · created 2026-08-02T01:08:38+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458804.80534 +/- 0.00064 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1274 +/- 0.0093
Transit depth (Rp/Rs)^2	1.62 +/- 0.24 [%]
Orbital Inclination (inc)	80.82 +/- 2.44
Airmass coefficient 1 (a1)	1.188 +/- 0.015
Airmass coefficient 2 (a2)	-0.1599 +/- 0.003
Scatter in the residuals of the lightcurve fit is	1.24 %
Best Comparison Star	None
Optimal Aperture	3.86
Optimal Annulus	9.93
Transit Duration (day)	0.1193 +/- 0.007

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1274 ± 0.0093 vs 0.1178 (deviation 0.77σ)
Duration fit vs published	0.1193 d vs 0.125 d (deviation 0.61σ)
Mid-time O–C	1.2 min
Depth SNR	10.2
Residual scatter	1.24 %

Light curve

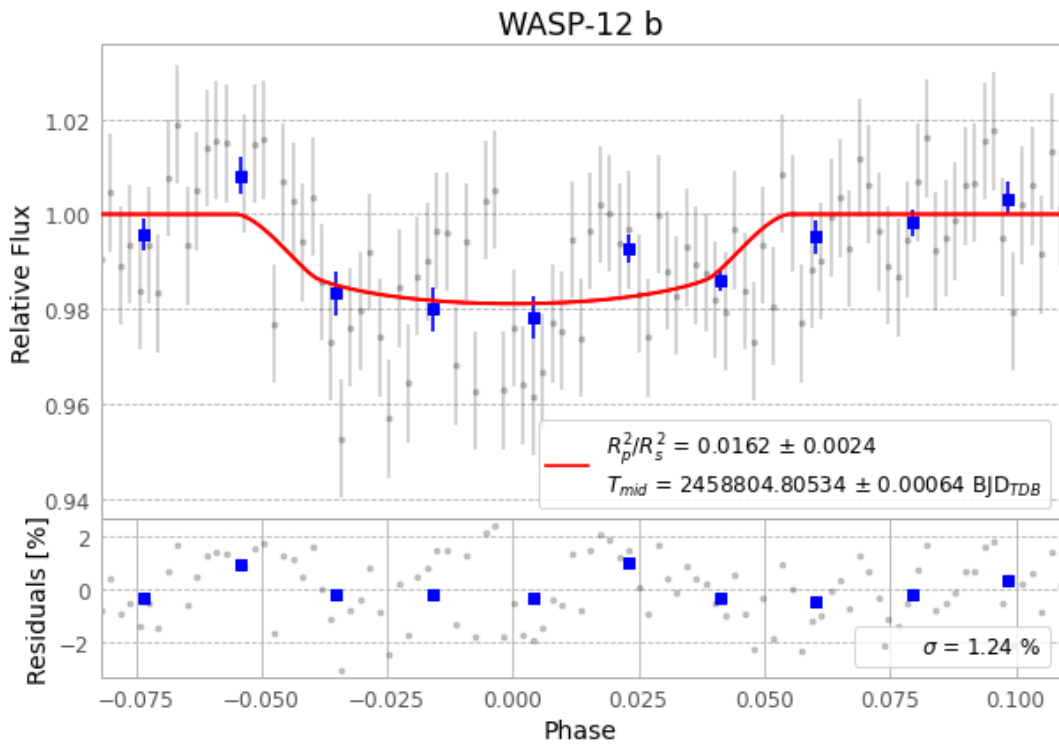


Figure 1 — FinalLightCurve_WASP-12 b_2019-11-16.png (SHA-256 9e675e982f7191f2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [283, 169], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7bf1ff246cd063cb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

101 FITS frames (66 MB), WASP-12191117050312.FITS → WASP-12191117100312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-191117.log (7d2b3da51b65...), FinalLightCurve_WASP-12 b_2019-11-16.png (9e675e982f71...), FinalLightCurve_WASP-12 b_2019-11-16.csv (4c37a3737b12...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 01:08Z.